Semantic Category Balance-Aware Involved Anti-Interference Network for Remote Sensing Semantic Segmentation

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***Abstract*— In recent years, semantic segmentation technology plays an important role in land resource management tasks. However, many classic semantic segmentation methods often fail to obtain satisfactory results for remote sensing images with a large amount of interference information. To improve this situation, we propose semantic category balance-aware involved anti-interference network (SCBANet). SCBANet has an encoder–decoder structure similar to DeeplabV3+. On this basis, we propose clustering-guided semantic decoupling module (CGSDM), consistency-based anti-interference feature extraction module (CAFEM), relevance-based anti-interference feature extraction module (RAFEM), and optional decoder module based on semantic category balance (ODMSCB) to improve the accuracy of semantic segmentation. CGSDM aims to obtain the information of different semantic categories through K-means clustering algorithm. CAFEM performs an average operation on the feature vectors in each semantic category to obtain semantic consistency information. RAFEM deeply excavates the information contained in each semantic category through the modeling method with self-attention mechanism as the core, making the relationship between pixels within each semantic category to be better understood by the model. ODMSCB classifies the feature map according to the balance of different semantic categories, so that different decoders can be applied to feature maps with different semantic category balance. These four parts complement each other, greatly improving the model's anti-interference ability while also enhancing the ability to handle category imbalance issue. We compared our method with several of the most advanced deep learning methods on the Vaihingen and Potsdam datasets. The final results demonstrate the superiority of our method.**

***Index Terms*— Remote sensing images, semantic category, semantic segmentation, strong anti-interference.**

I. Introduction

THE data required by traditional geography, environmental science and earth science research are obtained by ground monitoring stations or field visits, which have the defects of long cycle, long time-consuming, and high cost. However, remote sensing technology is a technology to obtain various information about ground objects from a long distance through sensors, which overcomes the shortcomings of traditional methods. Its main research object is ground object information, and its main content is to obtain ground object information for various applications, such as land change surveys, forest cover surveys, marine pollution surveys, and urban land surveys. It is of great significance to provide more useful information for geographical science research and development. In recent years, remote sensing technology has developed rapidly. Semantic segmentation based on remote sensing images has become a very important task in land resource management and road detection [2], [3], [4], [5].

Semantic segmentation refers to the problem of assigning semantic object categories to each pixel in a picture. Fully convolution network (FCN) [6] is the first classical semantic segmentation algorithm under deep learning. After that, many classical semantic segmentation methods have been proposed. The method based on the structure of the encoder and decoder [7], [8] solves the problem of aggregating semantic information and location information, which consists of a contracting path and an expanding path. The contracting path can extract abstract semantic features while the expanding path restores the location information. The method based on graph convolution network [9], [10] digs deep information of image by constructing convolution operator and pooling operator to achieve high-accuracy prediction results. Method based on multiscale contextual information [11] performs global pooling operation on feature maps, generates feature maps of different scales by setting different pooling rates, and finally concatenates them. In this way, local information and global information are fused, and finally, the prediction accuracy is improved. The method based on attention mechanism [12], [13], [14] has made outstanding contributions to solving the problem of how to obtain the relationship between long-distance context information in the spatial dimension and channel dimension. The model based on transformer structure [15], [16], [17] has demonstrated excellent ability in obtaining dependency relationships between pixels.

But there is one problem that deserves our attention. When we use the above model for semantic segmentation of remote sensing images, we usually cannot obtain good results. This is because remote sensing images have their own unique

Manuscript received 14 September 2023; accepted 13 October 2023. Date of publication 18 October 2023; date of current version 31 October 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFF0704000, in part by the National Natural Science Foundation of China under Grant 62172376 and Grant 62072418, and in part by the Central Government Guide Local Science and Technology Development Special Fund Project under Grant YDZX2022028. *(Jie Nie and Zhaoxin Wang are co-first authors.) (Corresponding author: Xinyue Liang.)*

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Digital Object Identifier 10.1109/TGRS.2023.3325327

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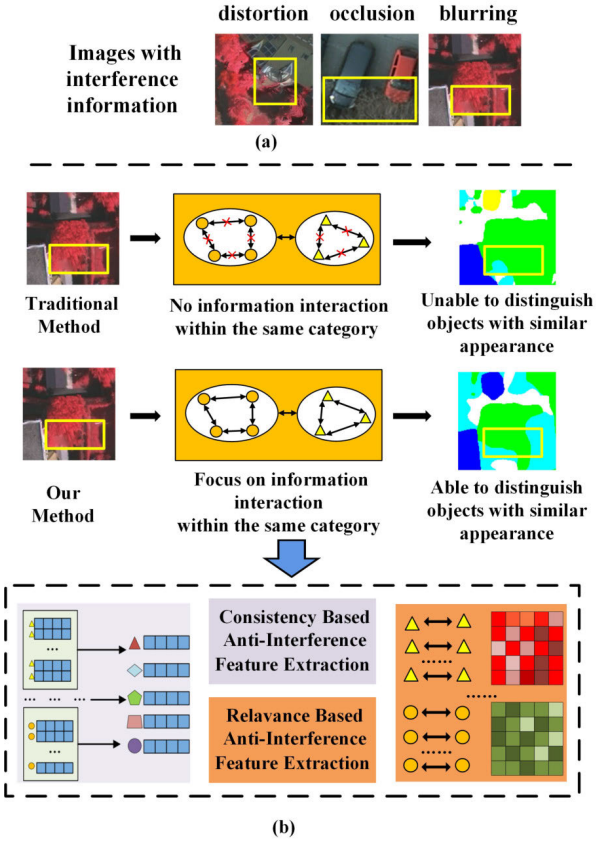


Fig. 1. Common types of interference information in remote sensing images in (a). (a) Distortion of objects, occlusion of objects, and blurring of objects. (b) Advantages of our method compared with traditional method in processing remote sensing images with interference information. Specifically, traditional method often has no interaction of the same category information, while focusing on interaction of the different category information. However, our method uses two modules, consistency-based anti-interference feature extraction and relevance-based anti-interference feature extraction, to complete the interaction of the same category information, ultimately achieving excellent semantic segmentation results. Each geometric shape (such as triangles, circles) represents a certain category.

characteristics. That is, remote sensing images contain a lot of interference information, such as object distortion, object occlusion, and object blurring [as shown in Fig. 1(a)]. Object distortion and object blurring often caused by the atmospheric scattering phenomenon or the sensor's view angle conversion is too fast. The state presented by the object will be different from normal situations, so it is difficult for the model to segment it correctly. Object occlusion requires the model to have a strong spatial analysis capability. For example, when branches cover cars, the model cannot generally divide the whole area into tree categories or car categories. When faced with remote sensing images containing a large amount of interference information, many classical models can only obtain the shallow information in the feature map and cannot mine the deep information contained in the feature map. This often leads to the deep learning model being affected by a large amount of interference information and finally achieves poor results. However, from Fig. 1(b), it can be seen that our proposed model has the ability to deal with interference information. Specifically, traditional method often has no interaction of the same category information, while focusing on interaction of the different category information. However, our method use two modules, consistency-based anti-interference feature extraction and relevance-based anti-interference feature extraction, to complete the interaction of the same category information, ultimately achieving excellent semantic segmentation results.

In addition, there are significant differences in category balance in remote sensing images. Classical models often do not have the ability to simultaneously process remote sensing images with different category balance.

To solve the above problems, we propose semantic category balance-aware involved anti-interference network (SCBANet). The core innovation of the network is to propose the concept of semantic category. Specifically, we assign each pixel on the feature map with its own semantic category and deeply mine the information contained in each semantic category, which enables the model to obtain more in-depth information, and ultimately greatly improves the model's ability to resist the above interference information. At the same time, we divide different remote sensing images into different categories with different levels of balance and pass them through different decoders. This operation aims to enable different decoders to process different types of remote sensing images in a targeted manner, ultimately enhancing the model's ability to handle large differences in category balance.

Specifically, our innovative content can be divided into four parts. The first is the clustering-guided semantic decoupling module (CGSDM). In this module, we use K-means clustering to assign each pixel on the feature map its own semantic category. At the same time, the module will judge the semantic category balance of the feature map based on certain rules, which lays the foundation for smooth operation in the optional decoder module based on semantic category. The second part is the consistency-based anti-interference feature extraction module (CAFEM). In this module, we perform average operation on the feature vectors in each semantic category to obtain semantic consistency information, thereby enabling the model to have a deeper understanding of the differences between each semantic category. The third part is the relevance-based anti-interference feature extraction module (RAFEM). In this module, we designed a semantic category relevance information enhancement method with the self-attention mechanism as the core, so that the relationship between pixels belonging to different semantic categories can be more clearly presented. The fourth part is the optional decoder module based on semantic category balance (ODMSCB). In this module, we make different feature maps enter different decoder according to the semantic category balance defined for the feature map in CGSDM. This allows different types of decoders to process different types of feature maps in a targeted manner, ultimately improving the semantic segmentation results. We compare our method with the previous method on the Vaihingen and Potsdam datasets. Experiments show that our method is superior to the state-of-the-art semantic segmentation models. Our main contributions are as follows.

1) We use the clustering algorithm to assign semantic category to each pixel on the feature map and propose SCBANet based on this operation, which ultimately

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improves the model's anti-interference ability and the ability to handle significant differences in category balance.

2) We propose CGSDM, CAFEM, RAFEM, and ODM-SCB. These four modules complement each other and ultimately improves the prediction accuracy of the model.
3) We validate the effectiveness of the proposed method on the Vaihingen and Potsdam datasets. Comparison and ablation experiments are performed to demonstrate the effectiveness of our proposed framework in remote sensing image segmentation tasks.

The remainder of this article is organized as follows. Section II presents the related works on the semantic segmentation of remote sensing images. Section III provides the details of the SCBANet. The corresponding experimental results and analysis are described in Section IV. Finally, we discuss the contributions of this article and conclude in Section V.

II. Related Work

A. Semantic Segmentation of Remote Sensing Images

Remote sensing images have become an important source of information for geographical science research. Remote sensing images can truly and vividly reflect the current situation of the distribution of ground objects, the relationship between ground objects or phenomena, and the mutual influence and change between ground objects. The introduction of remote sensing provides convenience and foundation for further in-depth study of regional comprehensive analysis and regional dynamic analysis of geography, as well as improving the practical application value and benefit of the results. So the semantic segmentation of remote sensing image is very necessary.

FCN [6] is the first classical semantic segmentation algorithm under deep learning. It replaces the full connection layer of the convolution neural network used for classification with deconvolution operation, which lays a foundation for the vigorous development of the semantic segmentation algorithm. Later neural networks are mostly based on FCN. At the same time, Ronneberger et al. [7], based on the idea of the FCN network model, proposed U-Net network, which has made brilliant achievements in ISBI competition in the year when it was proposed. The U-Net network model has been proved to be a very efficient network model through a large number of experiments.

Badrinarayanan et al. [8] put forward the concept of SegNet which is a new deep neural network. The network structure is constructed by the convolution layer. SegNet is divided into two parts. The left side is the feature extraction network at the encoder side, which extracts the feature map of multiple dimensions in the image. On the right is the decoder, which maps the features of multiple dimensions extracted by the encoder to a new feature space for pixel prediction to complete the task of image segmentation.

Zhao et al. [11] proposed a pyramid scene analysis network model, namely, pspnet. The structure of the model is like a pyramid. The size of the feature map at the lowest end of the feature extraction network is close to the original map, and the size of the feature map at the top is small. The feature maps of different sizes output in multiple stages are fused to obtain more comprehensive feature information. In the PSPNet network model, not only the main loss function but also the additional loss function is used.

DeeplabV1 [18] introduced the atrous convolution to solve the problem of pooled image receptive fields and added conditional random fields to optimize the segmentation results to obtain a more refined segmentation map. DeeplabV2 [19] has made further improvements on the basis of DeeplabV1: first, it has proposed the atrous spatial pyramid pool, which can obtain the context information of the target and image at multiple scales and second, instead of using the traditional conditional random fields, fully connected CRFs are further introduced to improve the positioning of segmented object edges. DeeplabV3 [20] has made some improvements on the cascade method, using the serial and parallel methods to achieve the purpose of multiscale object segmentation. In addition, the use of atrous convolution is discussed again, so as to obtain a larger receptive field and multiscale information. DeeplabV3+ takes DeeplabV3 as an encoder and adds a decoder to get a new segmentation model. This coding decoding structure can recover the edge information of the target image faster and better by reconstructing the space and further improve the prediction accuracy through atrous convolution.

Kampffmeyer et al. [21] improved FCN to apply it to the segmentation of small target objects (e.g., buildings, cars, trees) in remote sensing images. Guo et al. [22] proposed an integrated network for building identification, which overlays and fuses features extracted from networks such as AlexNet [23], VGGNet [24], GoogLeNet [25], and SqueezeNet [26] to obtain more expressive features. Pan et al. [27] used an U-Net network structure to extract buildings from high-resolution remote sensing images and incorporated the channel attention mechanism into it. MANet [28] proposed an interclass and intraclass region refinement method to reduce feature redundancy caused by fusion, WiCoNet [29] proposed a context transformer to embed context information in context branches, and TreeUNet [30] proposed a tree-shaped CNN structure to fuse multiscale features.

In recent years, models with transformer structure as the core have also achieved excellent results in semantic segmentation tasks. Vision Transformer [31] was first proposed to solve image classification tasks, and researchers in the field of semantic segmentation have discovered its enormous potential. So, researchers began using Vision Transformer for semantic segmentation tasks and proposed many excellent models based on this structure. Xie et al. [32] proposed the Segformer, which is a powerful and efficient semantic segmentation model. It combines transformers with lightweight multilayer perceptron decoders. Liu et al. [15] proposed Swin Transformer, which adopts a hierarchical structure and applies the shifted window concept, greatly improving the accuracy of semantic segmentation while reducing the computational complexity. Cao et al. [17] proposed SwinUnet, which combines the ideas

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of Swin Transformer and UNet to implement a U-shaped semantic segmentation model. Zhang et al. [16] proposed RegProxy. RegProxy presents a novel modeling that starts from interpreting the image as a tessellation of learnable regions, each of which has flexible geometrics and carries homogeneous semantics.

B. Semantic Segmentation With Decoupling Strategies

The Decoupling strategy is a very effective way to improve prediction accuracy in semantic segmentation tasks. Common decoupling strategies can be divided into the scale decoupling strategy and semantic decoupling strategy. The aforementioned PSPNet and DeeplabV3+ use the scale decoupling strategy, where PSPNet fuses feature maps of different scales to obtain multiscale information. The DeeplabV3+ uses the atrous spatial pyramid pooling module (ASPP) to extract the multiscale information contained in the feature maps. The scale decoupling strategy helps the above model achieve excellent semantic segmentation results, but the results obtained by this decoupling strategy when facing remote sensing images with a large amount of interference information are often not ideal. Therefore, researchers proposed the semantic decoupling strategy. The core idea of this strategy is to model multiple semantic objects (such as houses, trees, and cars) that exist in the image, thereby improving the intrasemantic consistency and intersemantic discriminability. This allows the deep neural network to have a deeper understanding of the relationships between pixels, which ultimately reduces the influence of interference information in remote sensing images to the deep neural network.

ACFNet [33] proposed the attentional class feature module, which can generate class centers according to the information in the feature map, and proposed the attentional class feature network based on this module. CDGCNet [34] proposes a classwise dynamic graph convolution module to adaptively propagate information and use this module to optimize coarse segmentation results. CCANet [35] proposed a novel class-constraint coarse-to-fine attentional network, which enables the formation of class information constraints to obtain explicit long-range context information. BGCNet [36] proposed the boundary-aware graph convolution module to propagate features within the object. CGFDN [37] proposed the feature decoupling module to encode the class co-occurrence relationships into the convolutional features.

These methods have achieved excellent results in the field of semantic segmentation of remote sensing images. However, there is still room for improvement in the way of mining the deep information contained in feature maps. So we propose the SCBANet.

III. Proposed Method

In this section, we present the proposed SCBANet devised for improving the semantic segmentation of remote sensing images. First, an overview of our network is given to introduce the motivation and architecture. After this, the proposed modules are described in detail. Finally, we explain the loss function used for network training.

A. Overview of the Proposed SCBANet

In Section I, we show object distortion, object occlusion and object blurring that often occur in remote sensing images. To deal with these interference information, and finally make the model obtain excellent semantic segmentation results in low-quality remote sensing images, we propose SCBANet.

Specifically, our network has four innovative contents: CGSDM, CAFEM, RAFEM, and ODMSCB. In the CGSDM, we use K-means clustering to assign each pixel on the feature map its own semantic category. At the same time, the module will judge the semantic category balance of the feature map based on certain rules, which lays the foundation for smooth operation in the optional decoder module based on semantic category. In the CAFEM, we perform average operation on the feature vectors in each semantic category to obtain semantic consistency information, thereby enabling the model to have a deeper understanding of the differences between each semantic category. In the RAFEM, we designed a semantic category relevance information enhancement method with the self-attention mechanism as the core, so that the relationship between pixels belonging to different semantic categories can be more clearly presented. In the ODMSCB, we make different feature maps enter different decoder according to the semantic category balance defined for the feature map in CGSDM. This allows different types of decoders to process different types of feature maps in a targeted manner, ultimately improving the semantic segmentation results.

The architecture of SCBANet is shown in Fig. 2. Given a remote sensing image, we first use the pretrained ResNet [38] and the ASPP to process it to generate the feature map X. Next, feature map X will enter CGSDM. In this module, we will perform the following operations: First, we will use clustering algorithm to cluster pixels on the feature map. At the same time, we will classify feature maps with varying degrees of category balance based on the clustering results. This decision result will help us determine which decoder in the ODMSCB the feature map will enter. After clustering algorithm, each pixel on the feature map X will be assigned to a semantic category. We assume that there are K semantic categories (K is equal to the number of categories of labeled data in the dataset) in total, and then we will obtain the feature map of K semantic categories, Next, these feature maps will enter the semantic consistency information extraction module and the semantic relevance information extraction module, respectively, to obtain the feature maps Y and Z. These three feature maps X, Y, and Z will be concatenated and enter the corresponding semantic category balance decoder to obtain the final segmentation result. Next, we will calculate the cross-entropy (CE) between the final segmentation result and the ground truth to obtain the loss value of the current epoch of training. Finally, we will use the Adam optimizer to optimize the SCBANet parameters and start the next epoch of training. Next, we will introduce each part in detail.

B. Clustering-Guided Semantic Decoupling Module

Since the operations of the optional decoder module based on semantic category, semantic consistency information

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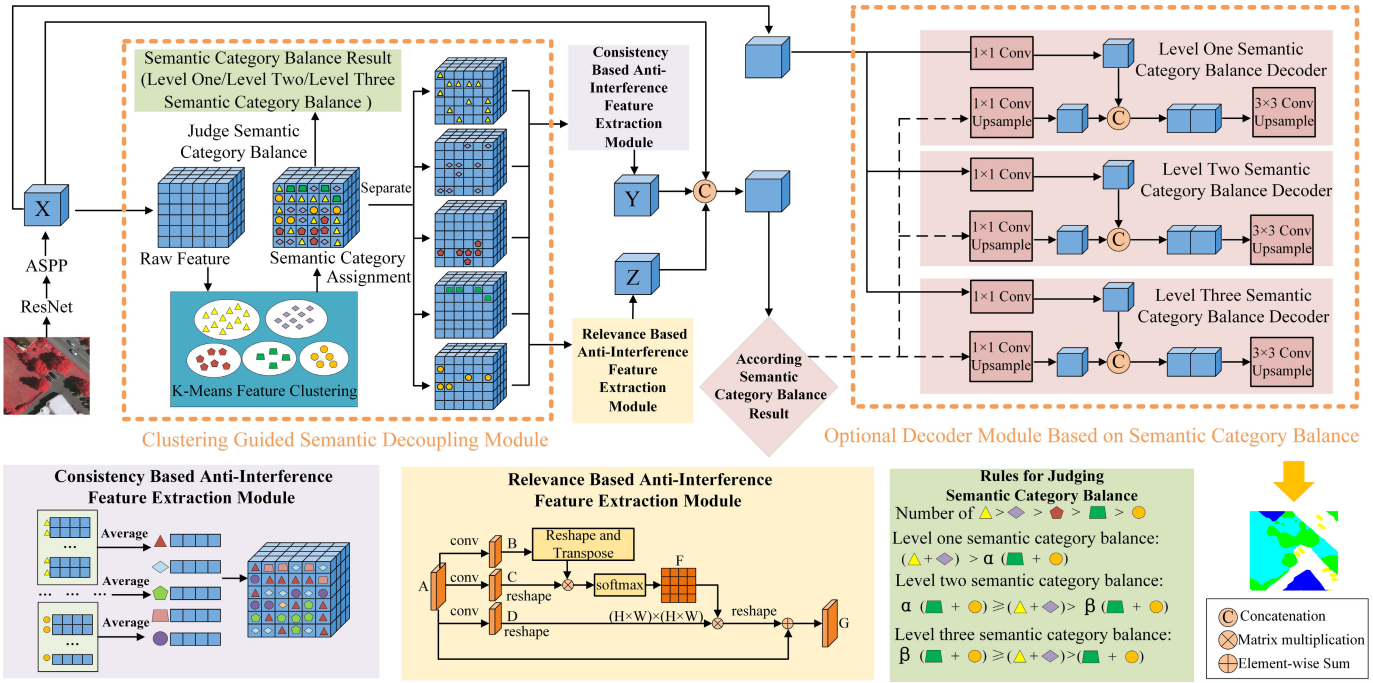


Fig. 2. Architecture of the proposed SCBANet is shown in Fig. 2. CGSDM, CAFEM, RAFEM, and ODMSCB are four innovation of this article. The three areas with mauve, yello, and green backgrounds at the bottom of Fig. 2 show the detailed structure of CAFEM, the detailed structure of RAFEM, and the rules for judging semantic category balance. Level one category balance decoder, level two semantic category balance decoder, and level three semantic category balance decoder in the right area constitute ODMSCB together.

extraction module, and semantic relevance information extraction module are based on different semantic category features, we will first introduce the process of semantic category assignment.

The raw remote sensing image is processed by the ResNet and the ASPP to obtain the feature map $X \in \mathbb{R}^{C\times H\times W}$. At this time, there are $H \times W$ pixels on the feature map, and each pixel is represented by a C-dimensional vector. Next, we perform K-means clustering on these $H \times W$ pixels. K-means algorithm is a classical clustering algorithm. Its core idea is that we first randomly set the centroids of K clusters, and then divide the samples into each cluster according to the nearest neighbor theory. Then, recalculate the centroid of each cluster until the change of the centroid is less than the given value. Specifically, for $H \times W$ pixel points represented by C-dimensional vector $L = \{L_1, \ldots, L_{H\times W}\}$, the objective of clustering is to minimize the squared error

$$E = \sum_{k=1}^{K} \sum_{L_i \in C_k} \|L_i - \mu_k\|_2^2 \tag{1}$$

where $C_k (k = 1, \ldots, K)$ represents the partition of cluster, and μ_k is the centroid of the cluster C_k. Through K-means clustering, we can obtain K different semantic clusters and assign a semantic category to each L_i.

In addition to implementing the function of assigning semantic categories to each pixel, CGSDM also divides different feature maps into feature maps with different semantic category balance degrees based on the semantic category assignment of pixel points. The specific division rules will be introduced in ODMSCB. In general, the operations in this module lay a solid foundation for the following operations.

C. Consistency-Based Anti-Interference Feature Extraction Module

The core idea of the CAFEM is to extract the representative characteristics of each semantic category. We refer to this characteristic as semantic consistency information, which can enable deep neural networks to have a deeper understanding of each semantic category, thereby achieving the effect of reducing the impact of interference information in remote sensing images on the model. The specific operation process is shown below. After the semantic category division process of feature map X, each pixel on it is assigned its own semantic category. Next, we generate K semantic category feature maps $M \in \mathbb{R}^{C\times H\times W}$ based on the feature map X assigned to the semantic category. The embedding vector corresponding to each pixel point is represented as follows:

$$M_{j(w,h)} = \begin{cases} X_{w,h}, & \text{if class}(w,h) = j \\ 0, & \text{else} \end{cases} \tag{2}$$

where $j \in [1, K]$ represents the semantic category to which the pixel is divided. $w \in [1, H]$ represents the number of rows of the pixel on the feature map. $h \in [1, H]$ represents the number of columns of the pixel on the feature map. (w, h) represents the position coordinate of the pixel. Class(w, h) represents the semantic category of the pixel. Next, we average the $H \times W$ embedding vectors in each feature map, and finally get K C-dimensional embedding vectors $P \in \mathbb{R}^C$, representing the representative features of each semantic category

$$P_j = \text{average}(M_{j(w,h)}) \tag{3}$$

where $j \in [1, K]$ represents the semantic category to which the pixel is divided. $w \in [1, H]$ represents the number of

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rows of the pixel on the feature map. $h \in [1, H]$ represents the number of columns of the pixel on the feature map. (w, h) represents the position coordinate of the pixel.

Finally, we will build semantic category representative characteristic feature map $Y \in \mathbb{R}^{C\times H\times W}$ based on these K embedding vectors

$$Y_{(w,h)} = P_j \quad \text{if class}(w, h) = j \tag{4}$$

where $j \in [1, K]$ represents the semantic category to which the pixel is divided. $w \in [1, H]$ represents the number of rows of the pixel on the feature map. $h \in [1, H]$ represents the number of columns of the pixel on the feature map. (w, h) represents the position coordinate of the pixel. class(w, h) represents the semantic category of the pixel.

D. Relevance-Based Anti-Interference Feature Extraction Module

Research by scientists in the field of human vision has shown that different parts of the human retina have different levels of information processing capabilities. To make reasonable use of the limited resources of visual information processing, humans need to select a specific part of the visual area and focus on it. Influenced by this idea, researchers in the field of computer vision have proposed attention mechanisms. Among them, the position attention module proposed in DANet [12] is a very classic self-attention mechanism, which has a strong ability to model the relationship between pixels. In the RAFEM, we obtained K semantic category feature maps $M_{j(w,h)} \in \mathbb{R}^{C\times H\times W}$. In this part, these feature maps will successively pass the self-attention mechanism to achieve the purpose of enhancing the connection between pixels of various semantic categories. The specific process is as follows.

Given feature map $A \in \mathbb{R}^{C\times H\times W}$, it will be fed in convolution layers to generate $B, C, D \in \mathbb{R}^{C\times H\times W}$. Then, reshape B and C to $\mathbb{R}^{C\times(H\times W)}$. After that matrix multiplication will be performed between C and the transpose of B. Next, the obtained result will be passed through the softmax layer to get the spatial attention graph F

$$F_{ji} = \frac{\exp(B_i \cdot C_j)}{\sum_{i=1}^{\mathrm{H\times W}} \exp(B_i \cdot C_j)} \tag{5}$$

where F_{ji} represents the influence degree of the ith position on the jth position. The more similar the feature representation of two locations, the greater the correlation between them. Then, matrix multiplication is performed on the feature map D and the spatial attention map F, the dimension of the obtained result will be reshaped to $\mathbb{R}^{C\times(H\times W)}$, multiplied by the learnable parameter γ, and the corresponding elements are added with the feature map A to obtain the final prediction map G

$$G_j = \gamma \sum_{i=1}^{\mathrm{H\times W}} (F_{ji} D_i) + A_j. \tag{6}$$

Finally, K feature maps that have passed the self-attention mechanism will be concatenated to generate semantic category attention mechanism feature map Z.

After feature maps of different semantic categories passing through the RAFEM, the deep neural network can better understand the relationship between pixels within each semantic category, thereby improving the resistance of the model to interference information existing in remote sensing images.

E. Optional Decoder Module Based on Semantic Category Balance

In the part of CGSDM, we assign the semantic category to each pixel on the feature map. Given feature map $X \in \mathbb{R}^{\text{Batch size}\times C\times H\times W}$, according to the number of pixels of each semantic category, we can determine the semantic category balance of the feature map. First, we count the total number of pixels of various semantic categories in a batch size, sort them from largest to smallest, and record them as $N_1, N_2, N_3, \ldots, N_K$. We will determine the semantic category balance of the feature map based on these numbers. The specific rules are as follows.

Level one semantic category balance

$$\begin{cases} \sum_{i=1}^{\frac{K-1}{2}} N_i > \alpha \sum_{i=\frac{K-1}{2}+2}^{K} N_i \\ \quad \text{if } K \text{ is an odd number} \\ \sum_{i=1}^{\frac{K}{2}} N_i > \alpha \sum_{i=\frac{K}{2}+1}^{K} N_i \\ \quad \text{if } K \text{ is an even number.} \end{cases} \tag{7}$$

Level two semantic category balance

$$\begin{cases} \alpha \sum_{i=\frac{K-1}{2}+2}^{K} N_i \geq \sum_{i=1}^{\frac{K-1}{2}} N_i > \beta \sum_{i=\frac{K-1}{2}+2}^{K} N_i \\ \quad \text{if } K \text{ is an odd number} \\ \alpha \sum_{i=\frac{K}{2}+1}^{K} N_i \geq \sum_{i=1}^{\frac{K}{2}} N_i > \beta \sum_{i=\frac{K}{2}+1}^{K} N_i \\ \quad \text{if } K \text{ is an even number.} \end{cases} \tag{8}$$

Level three semantic category balance

$$\begin{cases} \beta \sum_{i=\frac{K-1}{2}+2}^{K} N_i \geq \sum_{i=1}^{\frac{K-1}{2}} N_i > \sum_{i=\frac{K-1}{2}+2}^{K} N_i \\ \quad \text{if } K \text{ is an odd number} \\ \beta \sum_{i=\frac{K}{2}+1}^{K} N_i \geq \sum_{i=1}^{\frac{K}{2}} N_i > \sum_{i=\frac{K}{2}+1}^{K} N_i \\ \quad \text{if } K \text{ is an even number} \end{cases} \tag{9}$$

where α and β are the hyperparameters, which are set to 1.8 and 1.5, respectively in this article. K represents the number of categories to which pixels in the feature map are classified.

To make it easier for everyone to understand the above rules, we will take the graphical formula at the bottom right of Fig. 2 as an example to further illustrate the above rules.

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In this example, after the clustering algorithm, the pixels on the feature map will be divided into five categories, represented by triangles, diamonds, regular pentagons, trapezoids, and circles. We assume that the result obtained is that the number of pixels belonging to the triangle category is the highest, the number of pixels belonging to the diamond category is the second highest, the number of pixels belonging to the regular pentagon category is the third highest, the number of pixels belonging to the trapezoidal category is the fourth highest, and the number of pixels belonging to the circular category is the lowest. Next, we will add the number of pixels in the top half of the categories (triangles and diamonds), which is called the number of large category pixels. At the same time, we will add the number of pixels in the bottom half of the categories (trapezoids and circles), which is called the number of small category pixels. Due to the odd number of categories in this example, the middle ranked category (regular pentagons) will not be considered when calculating the number of large category pixels and the number of small category pixels. Next, we will determine the semantic category balance of the feature map. The specific rules are as follows: if the number of large category pixels is greater than α times the number of small category pixels, the semantic category balance of the feature map is level one. If the number of large category pixels is greater than β times the number of small large category pixels and less than or equal to α times the number of small category pixels, the semantic category balance of the feature map is level two. If the number of large category pixels is less than or equal to β times the number of small category pixels, then determine that the semantic category balance of the feature map is level three.

So far, each feature map has its own semantic category balance. Next, each feature map will enter the decoder corresponding to its semantic category balance to complete the final operation. Classic networks often use only one decoder to process feature maps, but one decoder cannot accurately process feature maps with different semantic category balance at the same time. The ODMSCB we designed enables different types of decoders to process different types of feature maps, which solves this problem very well.

F. Loss Function

To train the parameters in semantic category involved strong anti-interference network, a CE loss function is applied to measure the distributional difference between the generated results and the ground truth, which can be expressed as follows:

$$\mathcal{L}_{\mathrm{CE}} = \frac{1}{n}\sum_{i=1}^{n}(-y_i \log(p_i) - (1-y)_i \log(1-p_i)) \quad (10)$$

where n refers to all the remote sensing images and p_i is the prediction generated by the semantic category involved strong anti-interference network with a ground-truth label of y_i.

IV. Experimental Results and Analysis

A. Datasets

Our proposed method is evaluated on the Vaihingen and Potsdam datasets of the ISPRS 2-D semantic labeling benchmark. The Vaihingen dataset contains 33 tiles with an average size of 2494 × 2064 pixels. The Potsdam dataset contains 38 tiles with the size of 6000 × 6000 pixels. Pixels in both the datasets are divided into six categories, including impervious surface, building, low vegetation, tree, car, and background. For the Vaihingen dataset, we use 16 images for training and the remaining 17 images for testing. For the Potsdam dataset, we use 24 images for training and the remaining 14 images for testing. In addition, to more convincingly demonstrate the strong anti-interference ability of our proposed model, we constructed the P–V dataset. The P–V subset is derived from the Potsdam and Vaihingen datasets and established to verify the effectiveness of the proposed model in dealing with interference information. The spatial sizes of the data files are 256 × 256 pixels. The P–V subset has a total of 300 images, of which 150 images are used for training and the remaining 150 for testing. The P–V subset totally defines three types of interference according to the Potsdam and Vaihingen datasets, which are object distortion caused by atmospheric scattering, object occlusion caused by spatiotemporal interlaced objects, and object blurring caused by the high similarity of objects.

B. Evaluation Metrics

In our article, pixel accuracy, mean intersection over union, and mean F1 score are selected as evaluation indicators

$$\mathrm{PA} = \frac{n_{\mathrm{correct}}}{n_{\mathrm{total}}} \quad (11)$$

where n_{correct} represents the number of pixels correctly classified, and n_{total} represents the total number of pixels

$$\mathrm{Mean_iou} = \frac{1}{k}\sum_{i=1}^{k}\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \quad (12)$$

$$F1 = \frac{1}{k}\sum_{i=1}^{k}\frac{2\times\frac{\mathrm{TP}}{\mathrm{TP+FP}}\times\frac{\mathrm{TP}}{\mathrm{TP+FN}}}{\frac{\mathrm{TP}}{\mathrm{TP+FP}}+\frac{\mathrm{TP}}{\mathrm{TP+FN}}} \quad (13)$$

where k represents the total number of categories of pixels. TP, FP, and TN denote the true positive, false positive, and true negative, respectively.

C. Implementation Details

In this section, we mainly introduce the implementation details of the proposed method. To train our network, we crop the images into 1000 random patches with spatial sizes of 512 × 512. Random flipping or mirroring is performed for data augmentation to better train the network. All the models are implemented with PyTorch and the Adam optimizer with a 0.0003 learning rate. We set the batch size to 10 and train the model for 25 epochs. All the experiments are conducted on a server with one NVIDIA 2080Ti GPU.

D. Baselines

1) FCN [6]: FCN is a milestone achievement in the field of semantic segmentation. It has achieved the goal of semantic segmentation tasks in an end-to-end manner.

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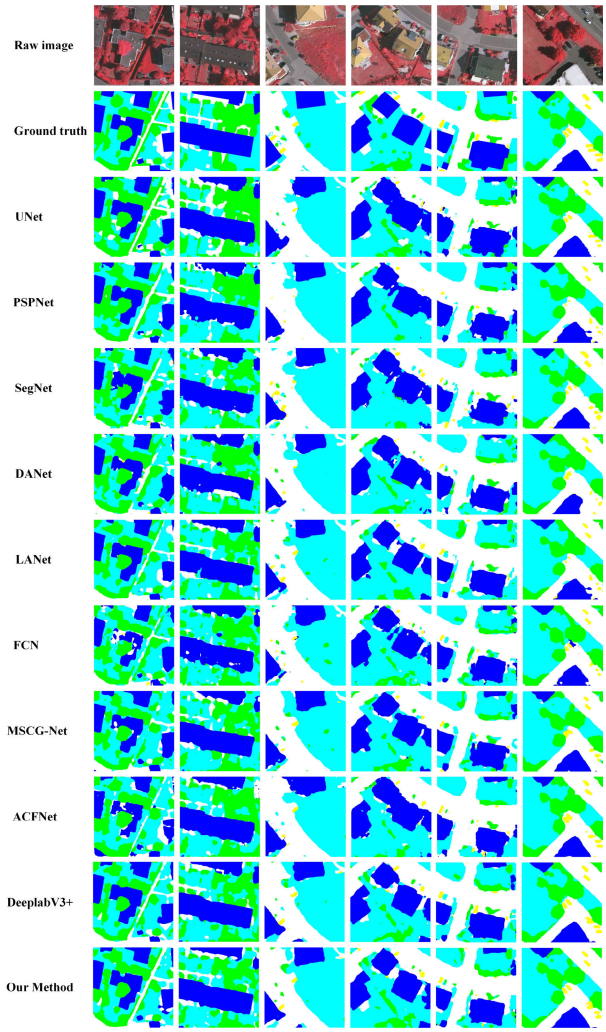


Fig. 3. Examples of semantic segmentation results on the Vaihingen test set.

2) ACFNet [33]: ACFNet is different from the general semantic segmentation deep learning model. It designs and uses context information from the spatial perspective, proposes the concept of class center, and extracts the global context from the perspective of classification.

3) UNet [7]: UNet is an encoder–decoder structure, which is composed of a contraction path and an expansion path. The contraction path can obtain abstract features, while the expansion path can restore the size of the segmentation map.

4) PSPNet [11]: PSPNet proposes a pyramid pooling module, which can aggregate context information of different scales, thus improving the ability of the model to obtain the information contained in the image.

5) Vision Transformer [39]: Vision Transformer was first proposed to solve image classification tasks, and researchers in the field of semantic segmentation have discovered its enormous potential and applied it to the semantic segmentation tasks.

6) Swin Transformer [15]: Swin Transformer adopts a hierarchical structure and applies the shifted window concept, greatly improving the accuracy of semantic segmentation while reducing the computational complexity.

7) Swin UNet [17]: SwinUNet combines the ideas of Swin Transformer and UNet to implement a U-shaped semantic segmentation model.

8) SegNet [8]: SegNet is also an encoder–decoder structure. Unlike U-Net, SegNet's encoder records the position index of the maximum value when performing max-pooling and uses the index to assist upsample in the decoder.

9) DANet [12]: DANet proposes the position attention module and channel attention module to explore the dependency between pixels.

10) LANet [40]: LANet uses the patch attention module to enhance the ability of the model to obtain context information.

11) MSCGNet [9]: MSCGNet is based on the self-constructing graph module, which uses the learnable potential variables to construct the basic graph directly from the input features without relying on the artificial prior knowledge graph.

12) CCANet [35]: CCANet proposed a novel class-constraint coarse-to-fine attentional network, which enables the formation of class information constraints to obtain explicit long-range context information.

13) DeeplabV3+ [1]: When using DeeplabV3+ for semantic segmentation tasks, good results can often be obtained. DeeplabV3+ adopts an encoder and decoder structure and adds ASPP to improve the semantic segmentation ability of the model.

14) CFPNet [41]: CFPNet proposed the multiscale convolution module aims to capture fine-grained multiscale context with different receptive fields, and the attention upsample module embeds the semantic information of high-level features into low-level features while maintaining spatial details.

15) LDCANet [42]: LDCANet proposed a dual-range context aggregation module to aggregate the local features and the global semantic context acquired by convolutions and self-attention respectively. Furthermore, a simple and lightweight decoder is used to combine information from different levels, in which a multilayer perceptron-based efficient linear block is proposed to yield a strong and efficient representation.

16) LoG-CAN [43]: LoG-CAN is a multiscale semantic segmentation network with a global class-aware module and local class-aware modules to remote sensing images. The global class-aware module captures the global representations of classwise context modeling to circumvent background interference; the local class-aware modules generate local class representations as intermediate aware elements.

E. Comparative Experiment

In this section, we carry out experiments to evaluate the effectiveness of our method. We adopt the same training and testing settings on the Vaihingen and Potsdam datasets. From the results of the comparative experiment, we can obtain the following analysis. Compared with the models using multiscale contextual feature extraction methods (DeeplabV3+), models using only ordinary feature extraction methods (FCN, SegNet, and UNet) have significantly lower semantic segmentation accuracy. In addition, DANet and LANet obtain better semantic segmentation accuracy by obtaining dependencies between pixels, while MSCG constructs a graph

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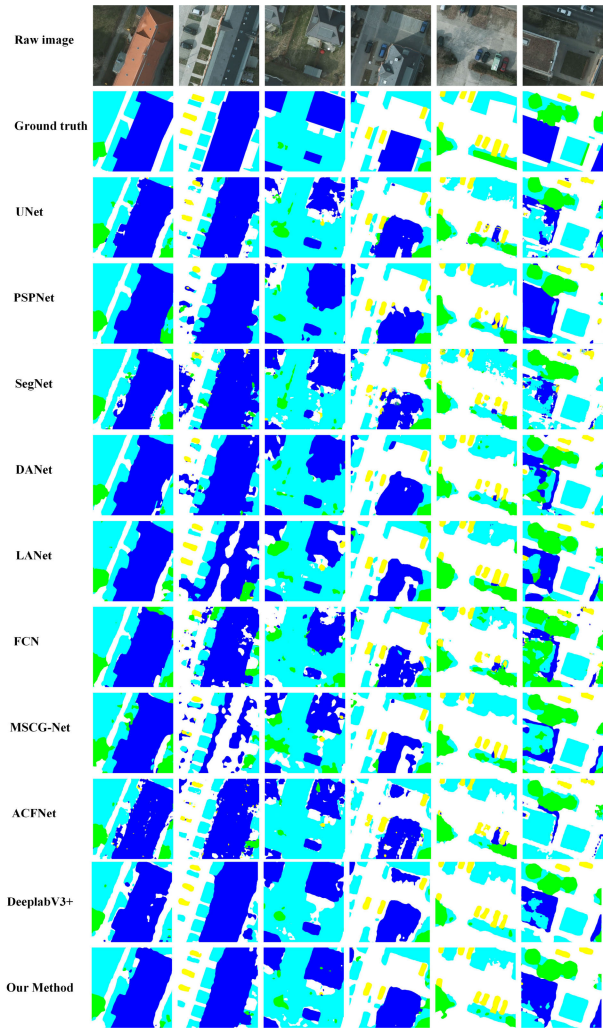


Fig. 4. Examples of semantic segmentation results on the Potsdam test set.

TABLE I
COMPARATIVE EXPERIMENTAL RESULTS ON THE VAIHINGEN TEST SET

Method	PA(%)	mIoU (%)	F1 score
FCN [6]	79.18	59.08	0.7297
ACFNet [33]	81.57	64.74	0.7806
UNet [7]	83.22	65.51	0.7815
PSPNet [11]	82.18	63.40	0.7636
Vision Transformer [39]	74.66	53.12	0.6806
Swin Transformer [15]	80.84	62.89	0.7651
SwinUNet [17]	79.16	57.90	0.7171
SegNet [8]	81.27	61.65	0.7487
DANet [12]	81.45	62.25	0.7549
LANet [40]	83.36	67.67	0.8017
MSCG-Net [9]	82.94	66.83	0.7951
CCANet [35]	83.99	67.32	0.7968
CFPNet [41]	83.47	68.31	0.8070
LDCANet [42]	83.72	67.38	0.7979
LoG-CAN [43]	82.77	68.53	0.8092
DeeplabV3+ [1]	84.30	69.38	0.8140
SCBANet	**85.07**	**70.76**	**0.8245**

TABLE II
COMPARATIVE EXPERIMENTAL RESULTS ON THE POTSDAM TEST SET

Method	PA(%)	mIoU (%)	F1 score
FCN [6]	72.45	53.28	0.6851
ACFNet [33]	79.65	61.05	0.7542
UNet [7]	78.62	62.40	0.7635
PSPNet [11]	79.57	62.59	0.7621
Vision Transformer [39]	66.13	46.33	0.6174
Swin Transformer [15]	74.27	54.88	0.7017
SwinUNet [17]	68.18	49.61	0.6552
SegNet [8]	77.22	57.30	0.7220
DANet [12]	77.75	60.46	0.7486
LANet [40]	78.56	63.40	0.7709
MSCG-Net [9]	79.23	63.88	0.7751
CCANet [35]	82.80	69.21	0.8158
CFPNet [41]	82.90	69.40	0.8168
LDCANet [42]	83.19	69.68	0.8197
LoG-CAN [43]	82.58	69.30	0.8158
DeeplabV3+ [1]	83.52	70.59	0.8256
SCBANet	**84.36**	**72.47**	**0.8377**

TABLE III
COMPARATIVE EXPERIMENTAL RESULTS ON THE P–V TEST SET

Method	PA(%)	mIoU (%)	F1 score
FCN [6]	70.04	53.27	0.6893
ACFNet [33]	75.33	58.86	0.7357
UNet [7]	78.86	65.32	0.7877
PSPNet [11]	78.75	66.17	0.7920
Vision Transformer [39]	64.70	44.16	0.6070
Swin Transformer [15]	72.35	54.32	0.6995
SwinUNet [17]	71.06	50.78	0.6687
SegNet [8]	77.45	63.73	0.7749
DANet [12]	77.99	63.31	0.7713
LANet [40]	78.02	65.23	0.7844
MSCG-Net [9]	76.28	60.91	0.7509
CCANet [35]	81.86	67.80	0.8063
CFPNet [41]	80.63	68.22	0.8082
LDCANet [42]	81.02	68.58	0.8105
LoG-CAN [43]	80.94	68.93	0.8136
DeeplabV3+ [1]	81.50	69.16	0.8152
SCBANet	**82.37**	**71.13**	**0.8288**

structure and improves the semantic segmentation ability of the model through graph reasoning. Overall, except for our model DeeplabV3+ achieved the best results. Quantitative results of the Vaihingen dataset are shown in Table I. Compared with DeeplabV3+, our method has increased by 0.77%, 1.38%, and 0.0105 in PA, Mean_iou, and $F1$ score, respectively. Quantitative results of the Potsdam dataset are shown in Table II. Compared with the best performing method, our method improves the pixel accuracy from 83.52% to 84.36%, the mean intersection over union from 70.59% to 72.47%, and the $F1$ score from 0.8256 to 0.8377. To more intuitively reflect the advantage of our method, we show the visualization diagram of the Vaihingen test set (Fig. 3) and Potsdam test set (Fig. 4). In addition, our proposed SCBANet also achieved the best results on the P–V dataset. Quantitative results are shown in Table III. Compared with DeeplabV3+, our method has increased by 0.87%, 1.97%, and 0.0136 in PA, Mean_iou, and $F1$ score, respectively.

F. Ablation Experiment of CAFEM, RAFEM, and ODMSCB

In this part, we perform experiments to verify the effectiveness of the CAFEM, RAFEM, and ODMSCB. Quantitative results of the Vaihingen dataset are shown in Table IV. CAFEM increased the pixel accuracy by 0.36%, the mean intersection over union by 0.48%, and the F1 score by 0.0041. RAFEM increased these indicators by 0.27%, 0.52%, and 0.0042, respectively. ODMSCB has improved 0.28%, 0.64%,

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TABLE IV
ABLATION EXPERIMENT OF CAFEM, RAFEM, AND ODMSCB RESULTS ON THE VAIHINGEN TEST SET

Method	PA(%)	mIoU(%)	F1 score
DeeplabV3+	84.30	69.38	0.8140
DeeplabV3+ +CAFEM	84.66	69.86	0.8181
DeeplabV3+ +RAFEM	84.57	69.90	0.8182
DeeplabV3+ +ODMSCB	84.58	70.02	0.8191
DeeplabV3+ +CAFEM+ RAFEM	84.84	70.38	0.8217
DeeplabV3+ +CAFEM+ ODMSCB	84.94	70.35	0.8215
DeeplabV3+ +RAFEM+ ODMSCB	84.82	70.37	0.8216
DeeplabV3+ +CAFEM+ RAFEM+ ODMSCB	**85.07**	**70.76**	**0.8245**

TABLE V
ABLATION EXPERIMENT RESULTS OF CAFEM, RAFEM, AND ODMSCB ON THE POTSDAM TEST SET

Method	PA(%)	mIoU(%)	F1 score
DeeplabV3+	83.52	70.59	0.8256
DeeplabV3+ +CAFEM	83.83	71.19	0.8297
DeeplabV3+ +RAFEM	83.81	71.09	0.8291
DeeplabV3+ +ODMSCB	83.92	71.15	0.8299
DeeplabV3+ +CAFEM+ RAFEM	84.24	71.61	0.8325
DeeplabV3+ +CAFEM+ ODMSCB	84.10	71.75	0.8331
DeeplabV3+ +RAFEM+ ODMSCB	84.06	72.03	0.8347
DeeplabV3+ +CAFEM+ RAFEM+ ODMSCB	**84.36**	**72.47**	**0.8377**

and 0.0051 in pixel accuracy, mean intersection over union, and F1 score, respectively. Table V shows the quantitative results of the Potsdam dataset. The addition of CAFEM improves the pixel accuracy from 83.52% to 83.83%, the mean intersection over union from 70.59% to 71.19%, and the F1 score from 0.8256 to 0.8297. With the addition of RAFEM, PA, Mean_iou, and F1 score have increased by 0.29%, 0.50%, and 0.0035, respectively. ODMSCB increased these indicators by 0.40%, 0.56%, and 0.0043, respectively. At the same time, it is worth noting that when we randomly take two of the three modules and add them to the network, we also achieved good results. These results are sufficient to demonstrate the validity of CAFEM, RAFEM, and ODMSCB.

G. Ablation Experiment of Clustering Method

In this part, we will discuss the clustering methods applied in CGSDM. K-means series clustering algorithms, hierarchical clustering algorithms, and density-based clustering algorithms are very classic clustering algorithms, and they each have advantages when processing different types of data. Considering that the number of categories that each pixel can be divided into in the semantic segmentation task is fixed, for example, the dataset used in this article has five categories of pixel points. Because density-based clustering algorithms can adaptively generate the number of categories, this type of clustering algorithm is not selected in this experiment. In this part of the ablation experiment, we selected four algorithms for comparison: K-means, MiniBatch K-means, agglomerative clustering, and birch. The K-means algorithm has been introduced previously. MiniBatch K-means is a clustering algorithm based on K-means, which divides all the data into multiple batches. This method effectively shortens the running time of the algorithm, but the prediction accuracy of the algorithm often decreases. Agglomerative clustering is a hierarchical clustering algorithm. Its execution process can be summarized as follows: First, treat each data as a class, and then continuously merge the data based on the similarity between the data until they reach the number of categories we specify. The full name of birch is balanced iterative reduction and clustering using hierarchies. This algorithm proposes two concepts, clustering feature and CF tree, which improve clustering speed and quality. From the results shown in Tables VI and VII, it can be seen that the K-means algorithm achieves the best results.

TABLE VI
ABLATION EXPERIMENT OF CLUSTERING METHOD ON THE VAIHINGEN TEST SET

Method	PA(%)	mIoU (%)	F1 score
MiniBatch K-Means	84.48	69.53	0.8153
Agglomerative Clustering	84.40	69.63	0.8165
Birch	84.52	69.70	0.8167
K-Means	**84.58**	**70.02**	**0.8191**

TABLE VII
ABLATION EXPERIMENT OF CLUSTERING METHOD ON THE POTSDAM TEST SET

Method	PA(%)	mIoU (%)	F1 score
MiniBatch K-Means	83.68	70.83	0.8274
Agglomerative Clustering	83.75	71.00	0.8288
Birch	83.87	70.95	0.8284
K-Means	**83.92**	**71.15**	**0.8299**

TABLE VIII
ABLATION EXPERIMENT OF HYPERPARAMETERS α AND β ON THE VAIHINGEN TEST SET

α	β	PA(%)	mIoU (%)	F1 score
2.0	1.5	84.45	69.67	0.8164
1.8	1.3	84.38	69.59	0.8160
2.0	1.3	84.48	69.57	0.8159
2.0	1.8	84.34	69.74	0.8174
1.8	**1.5**	**84.58**	**70.02**	**0.8191**

H. Ablation Experiment of Hyperparameter α and β

In this part, we will discuss the hyperparameters α and β mentioned in the optional decoder module based on semantic category. As mentioned earlier, feature maps are divided

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TABLE IX
ABLATION EXPERIMENT OF HYPERPARAMETERS α AND β ON THE POTSDAM TEST SET

α	β	PA(%)	mIoU (%)	F1 score
2.0	1.5	83.71	70.81	0.8271
1.8	1.3	83.75	70.93	0.8280
2.0	1.3	83.81	70.64	0.8267
2.0	1.8	83.61	71.08	0.8289
1.8	**1.5**	**83.92**	**71.15**	**0.8299**

into high semantic category balance feature maps, medium semantic category balance feature maps, and low semantic category balance feature maps based on the corresponding rules. Hyperparameters α and β appear in these rules. As we all know, the distribution of features in different datasets has a high probability of being different. Therefore, the rules for judging the semantic category balance of feature maps need to be formulated according to different datasets, and hyperparameters α and β play this role. From the results in Tables VIII and IX, it can be seen that both the datasets achieve the best results when α is 1.8 and β is 1.5.

V. CONCLUSION

This article proposes a novel SCBANet to deal with the semantic segmentation task of remote sensing images with significant differences in category balance and a large amount of interference information. CGSDM, CAFEM, RAFEM, and ODMSCB are the innovation of this article. These four modules operate the feature map based on the semantic category, explore the deep information in the feature map, and finally complete the semantic segmentation task of remote sensing image. In the future research, we will also explore other ways to mine the deep information in the feature map to better complete the semantic segmentation task of remote sensing images.

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